superposition of “deflected upward” and “deflected downward.” We measure which way it went, for example, by detecting the deflected electron on a screen, which is hooked up to a dial with a pointer indicating Up or Down. An Everettian says that the pointer is a big macroscopic object that quickly becomes entangled with the environment, leading to decoherence and branching of the wave function. GRW can’t appeal to such a process, but something related happens.

It’s not that the original electron spontaneously collapses; we would have to wait for millions of years for that to become a likely event. But the pointer in the apparatus contains something like 10^{24} electrons, protons, and neutrons. All of these particles are entangled in an obvious way: they are in different positions depending on whether the pointer indicates Up or Down. Even though it’s quite unlikely that any specific particle will undergo spontaneous collapse before we open the box, chances are extremely good that at least one of them will—that should happen roughly 10^8 times per second.

You might not be impressed, thinking that we wouldn’t even notice a tiny subset of particles becoming localized in a macroscopic pointer. But the magic of entanglement means that if the wave function of just one particle is spontaneously localized, the rest of the particles with which that one is entangled will come along with it. If somehow the pointer did manage to avoid any of its particles localizing for a certain period of time, enough for it to evolve into a macroscopic superposition of Up and Down, that superposition would instantly collapse as soon as just one of the particles did localize. The overall wave function goes very rapidly from describing an apparatus pointing in a superposition of two answers to one that is definitively one or the other. GRW theory manages to make operational and objective the classical/quantum split that partisans of the Copenhagen approach are forced to invoke. Classical behavior is seen in objects that contain so many particles that it becomes likely that the overall wave function will undergo a series of rapid collapses.

GRW theory has obvious advantages and disadvantages. The primary advantage is that it’s a well-posed, specific theory that addresses the measurement problem in a straightforward way. The multiple worlds of the Everett approach are eliminated by a series of truly unpredictable collapses. We are left with a world that maintains the successes of quantum theory in the microscopic realm, while exhibiting classical behavior macroscopically.